

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286250

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

WU; Kai et al.

BATTERY CELL, BATTERY, AND POWER CONSUMING DEVICE

Abstract

A battery cell, a battery, and a power consuming device are provided. The battery cell includes: an electrode assembly; an electrode terminal, electrically connected to the electrode assembly; a housing, configured to accommodate the electrode assembly, where the housing includes a wall portion and a fixing structure, and the fixing structure is fixedly arranged on the wall portion and configured to fix the electrode terminal; an insulating member, arranged between the fixing structure and the electrode terminal; and an insulating protection member, at least partially covering an outer surface of the fixing structure, where the insulating protection member includes an insertion portion, and at least a part of the insertion portion is inserted between the fixing structure and the insulating member.

Inventors: WU; Kai (Ningde, CN), Huang; Shoujun (Ningde, CN), Chen; Xinxiang (Ningde, CN), Jiang; Xiangwei (Ningde, CN), Zheng; Yulian (Ningde, CN)

Applicant: CONTEMPORARY AMPEREX TECHNOLOGY CO., LIMITED (Ningde, CN)

Family ID: 1000008669163

Assignee: CONTEMPORARY AMPEREX TECHNOLOGY CO., LIMITED (Ningde, CN)

Appl. No.: 19/213010

Filed: May 20, 2025

Foreign Application Priority Data

CN

202310395215.7

Apr. 13, 2023

Related U.S. Application Data

parent WO continuation PCT/CN2023/117647 20230908 PENDING child US 19213010

Publication Classification

Int. Cl.: H01M50/567 (20210101); H01M50/188 (20210101); H01M50/586 (20210101)

U.S. Cl.:

CPC H01M50/567 (20210101); H01M50/188 (20210101); H01M50/586 (20210101);

Background/Summary

[0001] This application is a continuation of International application PCT/CN2023/117647 filed on Sep. 8, 2023 that claims priority to Chinese Patent Application No. 202310395215.7, filed Apr. 13, 2023. The content of these applications is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the field of battery technologies, and more specifically, to a battery cell, a battery, and a power consuming device.

BACKGROUND

[0003] Energy conservation and emission reduction are keys to sustainable development of the automobile industry, and due to advantages of energy conservation and environmental protection, electric vehicles become an important part of the sustainable development of the automobile industry. For the electric vehicles, the battery technology is also an important factor related to development of the electric vehicles.

[0004] An electrode terminal is arranged on a wall portion of a battery cell, to output electrical energy of the battery cell or charge the battery cell. At present, an insulating member is arranged in a mounting hole of the electrode terminal on a housing, to insulate the housing from the electrode terminal. However, in this structure, a creepage distance between the electrode terminal and the housing is small, and short circuit is likely to occur under the influence of an external conductive member.

SUMMARY

[0005] An objective of the embodiments of the present application is to provide a battery cell, a battery, and a power consuming device, to resolve a problem of being prone to short circuit under the influence of an external conductive member due to a small creepage distance between an electrode terminal and a housing in the related art.

[0006] According to a first aspect, an embodiment of the present application provides a battery cell, including: [0007] an electrode assembly; [0008] an electrode terminal, electrically connected to the electrode assembly; [0009] a housing, configured to accommodate the electrode assembly, where the housing includes a wall portion and a fixing structure, and the fixing structure is fixedly arranged on the wall portion and configured to fix the electrode terminal; [0010] an insulating member, arranged between the fixing structure and the electrode terminal; and [0011] an insulating protection member, at least partially covering an outer surface of the fixing structure, where the insulating protection member includes an insertion portion, and at least a part of the insertion portion is inserted between the fixing structure and the insulating member.

[0012] In the technical solution of the embodiments of the present application, the insulating protection member is arranged to cover the outer surface of the fixing structure, so that the outer surface of the fixing structure is insulated and protected, and the insulation performance of the outer surface of the fixing structure is improved, thereby reducing the influence of an external conductive member; and the insertion portion is inserted between the fixing structure and the insulating member, so that a creepage distance between the electrode terminal and the fixing

structure is increased, and the external conductive member can be better prevented from being inserted between the fixing structure and the electrode terminal, thereby improving the insulating safety and reducing a short circuit risk.

[0013] In some embodiments, the fixing structure includes a first fixing structure and a second fixing structure, at least one of the first fixing structure and the second fixing structure is fixedly connected to the wall portion, the first fixing structure is located on an inner side of the second fixing structure away from the housing in a thickness direction of the wall portion, the first fixing structure cooperates with the second fixing structure to clamp and fix the electrode terminal, at least a part of the insulating protection member covers an outer surface of the first fixing structure, and at least a part of the insertion portion is inserted between the first fixing structure and the insulating member.

[0014] The first fixing structure and the second fixing structure are arranged to clamp the electrode terminal through cooperation, so that mounting and fixing of the electrode terminal are facilitated, thereby facilitating assembly of the electrode terminal.

[0015] In some embodiments, the first fixing structure includes a connection portion and a clamping portion; in the thickness direction of the wall portion, the clamping portion and the second fixing structure are arranged opposite to each other, and the connection portion is located between the clamping portion and the second fixing structure; [0016] the insulating member includes a first insulating portion and a second insulating portion, at least a part of the first insulating portion is located between the clamping portion and the electrode terminal in the thickness direction of the wall portion, at least a part of the second insulating portion is located between the clamping portion and the electrode terminal in a first direction, and the first direction is a direction pointing from the clamping portion to the electrode terminal and perpendicular to the thickness direction of the wall portion; and [0017] at least a part of the insertion portion is located between the first insulating portion and the clamping portion.

[0018] The clamping portion is arranged to clamp and fix the electrode terminal through cooperation with the second fixing structure, and the connection portion is arranged to support the clamping portion on the wall portion for ease of fixing of the clamping portion; and the insulating member is provided with the first insulating portion and the second insulating portion, so that the clamping portion and the electrode terminal are better insulated and separated from each other.

[0019] In some embodiments, the insertion portion is provided with a first engagement portion, a second engagement portion is arranged between the first fixing structure and the insulating member, and the first engagement portion is in engagement fit with the second engagement portion.

[0020] The first engagement portion is arranged on the insertion portion, the second engagement portion is arranged between the first fixing structure and the insulating member, and the insertion portion is fixed through engagement fit between the first engagement portion and the second engagement portion, so that the insertion portion can be better prevented from falling off, and the insulating protection member and the first fixing structure can be better fitted and fixed.

[0021] In some embodiments, the first engagement portion includes a limiting protrusion, the second engagement portion is arranged on at least one of the first fixing structure and the insulating member, the second engagement portion includes a limiting groove, and at least a part of the limiting protrusion is accommodated in the limiting groove.

[0022] The limiting groove is provided on at least one of the first fixing structure and the insulating member, the limiting protrusion is arranged on the insertion portion, and the limiting protrusion is accommodated through the limiting groove, so that the insertion portion is clamped and fixed, and a structure is simple and easy to process and manufacture.

[0023] In some embodiments, the first engagement portion includes a limiting protrusion, the second engagement portion includes an accommodating gap formed between the first fixing structure and the second insulating portion, and at least a part of the limiting protrusion is accommodated in the accommodating gap.

[0024] The accommodating gap is provided between the first fixing structure and the second insulating portion, the limiting protrusion is arranged on the insertion portion, and the limiting protrusion is accommodated through the accommodating gap, so that the insertion portion is clamped and positioned, and a structure is simple and easy to process and manufacture.

[0025] In some embodiments, the clamping portion has a first end surface facing the first insulating portion, the clamping portion has a second end surface facing the second insulating portion, the first end surface and the second end surface are connected through a first transition surface, the first transition surface is a rounded surface or a chamfered surface, and the first transition surface and the second insulating portion are spaced apart to form the accommodating gap.

[0026] The first transition surface is arranged between the first end surface and the second end surface of the clamping portion, and the first transition surface is spaced apart from the second insulating portion to form the accommodating gap, so that the limiting protrusion is accommodated and positioned, and a structure is simple and easy to process and manufacture.

[0027] In some embodiments, the insulating protection member includes a first coverage portion covering an outer surface of the clamping portion, and the insertion portion is connected to the first coverage portion.

[0028] The first coverage portion is arranged to cover the outer surface of the clamping portion, and the insertion portion is connected to the first coverage portion, so that the outer surface of the clamping portion and a surface close to the electrode terminal are well insulated and protected, thereby improving the insulation performance.

[0029] In some embodiments, the first coverage portion is bonded to the clamping portion.

[0030] The first coverage portion is bonded to the clamping portion, so that the clamping portion is insulated and protected, and the connection and fixing between the first coverage portion and the clamping portion are facilitated, thereby facilitating connection and fixing between the insulating protection member and the first fixing structure.

[0031] In some embodiments, the insulating protection member further includes a second coverage portion attached against an outer surface of the connection portion, and the second coverage portion is connected to the first coverage portion.

[0032] The second coverage portion is arranged and attached against the outer surface of the connection portion, so that the connection portion can be insulated and protected, thereby better improving the insulating protection performance.

[0033] In some embodiments, a first clamping protrusion is arranged on the second coverage portion, and a first clamping groove that is engaged with the first clamping protrusion is provided on the connection portion; and/or [0034] a first engagement groove is provided on the second coverage portion, and a first engagement protrusion that is engaged with the first engagement groove is arranged on the connection portion.

[0035] The first clamping protrusion is arranged on the second coverage portion, and the first clamping groove is provided on the connection portion, so that the second coverage portion can be fixedly connected to the connection portion through engagement between the first clamping protrusion and the first clamping groove, and the second coverage portion is further engaged with the insertion portion, to fix the insulating protection member on the first fixing structure, thereby facilitating assembly and fixing.

[0036] The first engagement groove is provided on the second coverage portion, and the first engagement protrusion is arranged on the connection portion, so that the second coverage portion can be fixedly connected to the connection portion through engagement between the first engagement protrusion and the first engagement groove, and the second coverage portion is further engaged with the insertion portion, to fix the insulating protection member on the first fixing structure, thereby facilitating assembly and fixing.

[0037] In some embodiments, an angle between the second coverage portion and the first coverage portion is an acute angle.

[0038] The angle between the second coverage portion and the first coverage portion is an acute angle, so that the second coverage portion can cooperate with the insertion portion to clamp the first fixing structure, and the insulating protection member is further connected to the first fixing structure, thereby facilitating assembly.

[0039] In some embodiments, a surface of the second coverage portion close to the connection portion has a non-sticking characteristic.

[0040] The surface of the second coverage portion close to the connection portion has a non-sticking characteristic, so that engagement connection between the second coverage portion and the connection portion is facilitated, thereby facilitating assembly.

[0041] In some embodiments, the insulating protection member further includes an extension portion covering an outer surface of the wall portion, and the extension portion is connected to the second coverage portion.

[0042] The extension portion is arranged to cover the outer surface of the wall portion, so that an insulating protection effect can be better achieved.

[0043] In some embodiments, the first coverage portion, the second coverage portion, the extension portion, and the insertion portion are integrally formed and arranged.

[0044] The first coverage portion, the second coverage portion, the extension portion, and the insertion portion are integrally formed, so that processing and manufacturing of the insulating protection member are facilitated.

[0045] In some embodiments, the first fixing structure and the wall portion are an integrally formed structure; and/or [0046] the second fixing structure and the wall portion are an integrally formed structure; and/or [0047] the first fixing structure and the second fixing structure are an integrally formed structure.

[0048] The first fixing structure and the wall portion are integrally formed, so that connection and fixing between the first fixing structure and the wall portion are facilitated, and processing and manufacturing are also facilitated.

[0049] The second fixing structure and the wall portion are integrally formed, so that connection and fixing between the second fixing structure and the wall portion are facilitated, and processing and manufacturing are also facilitated.

[0050] The first fixing structure and the second fixing structure are integrally formed, so that connection and fixing between the first fixing structure and the second fixing structure are facilitated, and processing and manufacturing are also facilitated.

[0051] In some embodiments, an insulating sealing member is further arranged between the fixing structure and the electrode terminal.

[0052] The insulating sealing member is arranged between the fixing structure and the electrode terminal, so that the sealing performance is improved, and the insulating protection performance between the fixing structure and the electrode terminal can be improved.

[0053] In some embodiments, the insertion portion has a non-sticking characteristic.

[0054] The insertion portion has a non-sticking characteristic, so that the insertion portion can be inserted between the fixing structure and the insulating member, thereby facilitating assembly, and the insulating protection member can better fit and cover the outer surface of the fixing structure.

[0055] In some embodiments, the housing includes a shell and an end cap, one end of the shell has an opening, the end cap covers the opening, the shell includes a side wall and a bottom wall, the side wall is annularly disposed on an outer side of the electrode assembly, and the bottom wall and the opening are arranged opposite to each other; and the wall portion is at least a partial region of the end cap, or the wall portion is at least a partial region of the bottom wall, or the wall portion is at least a partial region of the side wall.

[0056] At least a partial region of the end cap is used as the wall portion, or at least a partial region of the bottom wall is used as the wall portion, or at least a partial region of the side wall is used as the wall portion, so that a layout design of the electrode terminal is facilitated.

[0057] According to a second aspect, an embodiment of the present application provides a battery, including the battery cell according to any one of the foregoing embodiments.

[0058] According to a third aspect, an embodiment of the present application provides a power consuming device, including the battery according to the foregoing embodiment.

[0059] The foregoing description only refers to an overview of the technical solutions of the present application. In order to understand the technical means of the present application more clearly, the present application can be implemented according to the content of this specification. In addition, to make the above-mentioned and other objectives, features, and advantages of the present application more comprehensible, specific implementations of the present application are listed below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0060] To describe the technical solutions of the embodiments of the present application more clearly, the accompanying drawings required for the embodiments or exemplary technical description will be briefly described below. Apparently, the accompanying drawings described below are only some embodiments of the present application. For a person of ordinary skill in the art, other accompanying drawings can be further obtained according to the accompanying drawings without creative efforts.

[0061] FIG. 1 is a schematic structural diagram of a vehicle according to some embodiments of the present application;

[0062] FIG. 2 is a schematic exploded structural view of a battery according to some embodiments of the present application;

[0063] FIG. 3 is a schematic exploded structural view of a battery cell according to some embodiments of the present application;

[0064] FIG. 4 is a schematic top structural view of mounting a wall portion of an electrode terminal on a housing according to some embodiments of the present application;

[0065] FIG. 5 is a schematic cross-sectional structural view along a line A-A in FIG. 4;

[0066] FIG. 6 is an enlarged view of a part B in FIG. 5;

[0067] FIG. 7 is an enlarged view of a part C in FIG. 6;

[0068] FIG. 8 is a schematic top structural view of a corresponding part of a second engagement portion in a battery cell according to some other embodiments of the present application;

[0069] FIG. 9 is a schematic top structural view of a corresponding part of a second engagement portion in a battery cell according to still some other embodiments of the present application;

[0070] FIG. 10 is a schematic top structural view of a corresponding part of a second engagement portion in a battery cell according to yet some other embodiments of the present application;

[0071] FIG. 11 is a schematic cross-sectional structural view of a connection part between an electrode terminal and a wall portion in a battery cell according to some other embodiments of the present application;

[0072] FIG. 12 is an enlarged view of a part D in FIG. 11;

[0073] FIG. 13 is a schematic cross-sectional structural view of a connection part between an electrode terminal and a wall portion in a battery cell according to still some other embodiments of the present application;

[0074] FIG. 14 is a schematic cross-sectional structural view of a connection part between an electrode terminal and a wall portion in a battery cell according to still some other embodiments of the present application;

[0075] FIG. 15 is an enlarged view of a part E in FIG. 14;

[0076] FIG. 16 is a schematic cross-sectional structural view of a connection part between an

electrode terminal and a wall portion in a battery cell according to yet some other embodiments of the present application;

[0077] FIG. **17** is a schematic cross-sectional structural view of a connection part between an electrode terminal and a wall portion in a battery cell according to yet some other embodiments of the present application;

[0078] FIG. **18** is a schematic cross-sectional structural view of a connection part between an electrode terminal and a wall portion in a battery cell according to some other embodiments of the present application;

[0079] FIG. **19** is an enlarged view of a part F in FIG. **18**;

[0080] FIG. **20** is a schematic structural diagram of connection between an extension portion and a wall portion in a battery cell according to some other embodiments of the present application;

[0081] FIG. **21** is a schematic cross-sectional structural view of a connection part between an electrode terminal and a wall portion in a battery cell according to yet some other embodiments of the present application; and

[0082] FIG. **22** is an enlarged view of a part G in FIG. **21**.

MAIN REFERENCE NUMERALS IN THE FIGURES

[0083] **1000**-Vehicle; **1001**-Battery; **1002**-Controller; **1003**-Motor; [0084] **100**-Box; **101**-First part; **102**-Second part; [0085] **200**-Battery cell; [0086] **21**-Electrode assembly; **211**-Tab; **22**-Adapting piece; [0087] **30**-Housing; **31**-Shell; **310**-Opening; **311**-Side wall; **312**-Bottom wall; **32**-[0088] End cap; **301**-Wall portion; **3011**-Second clamping groove; **3012**-Second engagement protrusion; [0089] **41**-Electrode terminal; **411**-Flange portion; [0090] **42**-Insulating member; **421**-First insulating portion; **422**-Second insulating portion; [0091] **43**-Insulating sealing member; [0092] **50**-Fixing structure; **51**-First fixing structure; **511**-Clamping portion; **5111**-First end surface; **5112**-Second end surface; **5113**-First transition surface; **51131**-Rounded surface; **51132**-Chamfered surface; **512**-Connection portion; **5121**-First clamping groove; **5122**-First engagement protrusion; **513**-Second transition surface; **514**-Third transition surface; **52**-Second fixing structure; [0093] **60**-Insulating protection member; **61**-First coverage portion; **62**-Insertion portion; **63**-Second coverage portion; **631**-First clamping protrusion; **632**-First engagement groove; **64**-First transition portion; **65**-Second transition portion; **66**-Extension portion; **661**-Second clamping protrusion; **662**-Second engagement groove; [0094] **70**-First engagement portion; **71**-Limiting protrusion; **711**-Bump structure; **712**-Hook; [0095] **80**-Second engagement portion; **81**-Accommodating gap; **82**-Limiting groove; **821**-Recessed groove; **822**-Clamping slot; [0096] **91**-First adhesive layer; and **92**-Second adhesive layer.

DETAILED DESCRIPTION

[0097] To make the technical problems to be resolved by the present application, the technical solutions, and the beneficial effects more comprehensible, the present application is further described below in detail with reference to the accompanying drawings and the embodiments. It should be understood that, the specific embodiments described herein are only used for describing the present application and are not intended to limit the present application.

[0098] Unless otherwise defined, all technical and scientific terms used in this specification have same meanings as those commonly understood by a person skilled in the art to which the present application belongs. The terms used in this specification are only intended to describe specific embodiments, and are not intended to limit the present application. The terms “include”, “comprise”, and any variants thereof in the specification and claims of the present application and the foregoing accompanying drawings are intended to cover non-exclusive inclusion.

[0099] In the description according to the embodiments of the present application, the technical terms “first”, “second”, and the like are only used to distinguish different objects, and should not be understood as indicating or implying relative importance or implicitly indicating a quantity, a specific order, or a primary and secondary relationship of indicated technical features. Therefore, a feature defined by “first” or “second” can explicitly or implicitly include one or more of the

features.

[0100] “Embodiment” mentioned in the specification means that particular features, structures, or characteristics described with reference to the embodiment may be included in at least one embodiment of the present application. The term appearing at different positions of the specification may not refer to the same embodiment or an independent or alternative embodiment that is mutually exclusive with another embodiment. A person skilled in the art explicitly or implicitly understands that the embodiments described in the specification may be combined with other embodiments in any suitable manner.

[0101] In the description of the embodiments of the present application, the term “and/or” describes only an association relationship for describing associated objects and represents that three relationships may exist. For example, A and/or B may represent the following three cases: Only A exists, both A and B exist, and only B exists. In addition, the character “/” in this specification generally indicates an “or” relationship between the associated objects.

[0102] In the description of the embodiments of the present application, the term “a plurality of” means two or more (including two). Similarly, “a plurality of groups” means two or more groups (including two groups), and “a plurality of pieces” means two or more pieces (including two pieces). Unless otherwise expressly specified and defined, “several” means one or more.

[0103] In the description of the embodiments of the present application, orientation or position relationships indicated by the technical terms such as “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise”, “counterclockwise”, “axial”, “radial”, and “circumferential” are based on orientation or position relationships shown in the accompanying drawings, and are only for the convenience of describing the present application and simplifying the description, rather than indicating or implying that the indicated device or element must have a specific orientation or must be configured and operated in a specific orientation. Therefore, such terms should not be construed as a limitation to the embodiments of the present application.

[0104] In the description of the embodiments of the present application, unless otherwise expressly specified and defined, the technical terms “mount”, “connect”, “connection”, “fix”, and the like should be understood in a broad sense. For example, the connection may be a fixed connection, a detachable connection, or an integral connection; or may be a mechanical connection or an electrical connection; or may be a direct connection, an indirect connection through an intermediary, or internal communication between two elements or a mutual interaction relationship between two elements. A person of ordinary skill in the art can understand specific meanings of the terms in the embodiments of the present application according to specific situations.

[0105] In the description of the embodiments of the present application, when an element is described as being “fixed to” or “arranged on” another element, the element may be directly located on the another element or indirectly located on the another element. When an element is described as being “connected to” another element, the element may be directly connected to the another element or indirectly connected to the another element.

[0106] In the description of the embodiments of the present application, unless otherwise clearly specified and defined, the technical term “adjacent” refers to being close in position. For example, for three components A1, A2, and B, if a distance between A1 and B is greater than a distance between A2 and B, when comparison is performed between A2 and A1, A2 is closer to B, that is, A2 is adjacent to B, or B is adjacent to A2. For another example, when there are a plurality of components C and the plurality of components C are respectively C1, C2, . . . , and CN, if a component C, for example, C2 is closer to the component B than other components C, B is adjacent to C2, or C2 is adjacent to B.

[0107] In the present application, a battery cell includes, but is not limited to, a lithium-ion secondary battery, a lithium-ion primary battery, a lithium-sulfur battery, a sodium-lithium-ion battery, a sodium-ion battery, or a magnesium-ion battery. The battery cell may be cylindrical, flat,

cuboid, or in other shapes, but is not limited thereto. Generally, the battery cell is divided into, but is not limited to, three types according to encapsulating methods: a cylindrical battery cell, a square battery cell, and a soft package battery cell.

[0108] A battery mentioned in the embodiments of the present application is a single physical module including one or more battery cells to provide a higher voltage and a higher capacity. For example, the battery mentioned in the present application may include a battery module, a battery pack, or the like. The battery generally includes a box for packaging the one or more battery cells. The box can prevent liquid or other foreign objects from affecting charging or discharging of the one or more battery cells. In some cases, the battery cells may alternatively be directly used, that is, the battery may alternatively not include the box.

[0109] In the battery, when there are a plurality of battery cells, the plurality of battery cells may be connected in series, parallel, or series-parallel. Series-parallel connection means that both series connection and parallel connection exist among the plurality of battery cells. The plurality of battery cells may be directly connected in series, parallel, or series-parallel together, and then a whole formed by the plurality of battery cells may be accommodated in the box. Certainly, the battery may alternatively be in the form of a plurality of battery modules formed by the plurality of battery cells that are first connected in series, parallel, or series-parallel, and the plurality of battery modules are then connected in series, parallel, or series-parallel to form a whole and accommodated in the box. The battery may also include other structures, for example, the battery may also include a bus component for achieving electrical connection between the plurality of battery cells.

[0110] The battery cell in the embodiments of the present application includes an electrode assembly and a housing, and the electrode assembly is mounted in the housing.

[0111] The electrode assembly is also referred to as a battery core, and the electrode assembly is formed by a positive electrode plate, a negative electrode plate, and a separator. The electrode assembly works mainly relying on movement of metal ions between the positive electrode plate and the negative electrode plate. The positive electrode plate includes a positive electrode current collector and a positive electrode active substance layer, the positive electrode active substance layer is coated on a surface of the positive electrode current collector, a part of the positive electrode current collector not coated with the positive electrode active substance layer protrudes out of a part of the positive electrode current collector already coated with the positive electrode active substance layer, and the part of the positive electrode current collector not coated with the positive electrode active substance layer is used as a positive electrode tab, or a metal conductor is welded to and led out from the positive electrode current collector as the positive electrode tab. By using a lithium-ion battery as an example, a material of the positive electrode current collector may be aluminum, and a positive electrode active substance may be lithium cobalt oxide, lithium iron phosphate, ternary lithium, lithium manganese oxide, or the like. The negative electrode plate includes a negative electrode current collector and a negative electrode active substance layer, the negative electrode active substance layer is coated on a surface of the negative electrode current collector, a part of the negative electrode current collector not coated with the negative electrode active substance layer protrudes out of a part of the negative electrode current collector already coated with the negative electrode active substance layer, and the part of the negative electrode current collector not coated with the negative electrode active substance layer is used as a negative electrode tab, or a metal conductor is welded to and led out from the negative electrode current collector as the negative electrode tab. A material of the negative electrode current collector may be copper, and a negative electrode active substance may be carbon, silicon, or the like. To ensure that no fusing occurs when a large current passes through, there are a plurality of positive electrode tabs and the positive electrode tabs are stacked, and there are a plurality of negative electrode tabs and the negative electrode tabs are stacked. It may be understood that, in the electrode assembly, there may be one positive electrode tab and there may also be one negative electrode tab. That is, two

groups of tabs are arranged on the electrode assembly, and each group includes at least one tab, where one group of tabs are positive electrode tabs, and the other group of tabs are negative electrode tabs.

[0112] A structure of the electrode assembly includes but is not limited to a wound structure or a laminated structure. In the wound structure, tabs are mostly welded to current collectors, and arrangement is then performed according to a sequence of positive electrode plate-separator-negative electrode plate-separator; and a cylindrical or square battery core is then formed through winding. In the laminated structure, tabs are mostly led out from current collectors, and positive electrode plates, negative electrode plates, and separators are then arranged according to a sequence of positive electrode plate-separator-negative electrode plate-separator, to laminate the positive electrode plates, the negative electrode plates, and the separators layer by layer to form a laminated battery core. The separator may be cut off and lamination is then directly performed by using a separating piece, or the separator is not cut off and lamination is performed in a Z-shaped lamination manner. The separator may be made of a material such as polypropylene (PP) or polyethylene (PE). The separator is an insulating film arranged between the positive electrode plate and the negative electrode plate and a main function thereof is: isolating a positive electrode from a negative electrode and stop electrons in the battery from passing through freely, to prevent short circuit, and enabling ions in an electrolyte to pass through freely between the positive electrode and negative electrode, to form a loop between the positive electrode and the negative electrode. The positive electrode plate and the negative electrode plate are collectively referred to as plates. The positive electrode tab and the negative electrode tab are collectively referred to as tabs.

[0113] An electrode terminal is arranged on a housing to be connected to the tabs of the electrode assembly, so as to output electrical energy of the battery cell or charge the battery cell.

[0114] To enable the battery cell to have a good structural strength, the housing is mostly made of a metal material, and to reduce a short circuit risk, an insulating member is generally arranged on a peripheral side of the electrode terminal on the housing, to insulate the housing from the electrode terminal. However, due to a volume limitation of the housing and the electrode terminal, the insulating member is only arranged between the electrode terminal and the housing, and a creepage distance between the electrode terminal and the housing is relatively small, which is prone to short circuit especially when an external conductive member is connected or conductive impurities fall on the housing during usage of the battery cell.

[0115] The creepage distance is a shortest distance measured along an insulating surface between two adjacent conductors or between a conductor and a surface of a housing of an adjacent motor.

[0116] Based on the foregoing concern, to resolve the problem of being prone to short circuit under the influence of an external conductive member caused by a small creepage distance between the housing and the electrode terminal, an embodiment of the present application provides a battery cell in which an insulating protection member is arranged to cover an outer surface of a fixing structure on a wall portion of a housing, so that the outer surface of the fixing structure is insulated and protected, and the isolation performance of the outer surface of the fixing structure is improved, thereby reducing the influence of the external conductive member; and an insertion portion is inserted between the fixing structure and an insulating member, so that a creepage distance between an electrode terminal and the fixing structure is increased, and the external conductive member can be better prevented from being inserted between the fixing structure and the electrode terminal, thereby improving the insulating safety and reducing a short circuit risk.

[0117] The battery cell disclosed in this embodiment of the present application may be used in a power consuming device using a battery as a power supply or various energy storage systems using a battery as an energy storage element, for example, used in an energy storage power supply system such as a hydraulic, thermal power, wind power, or solar power station. The power consuming device may be, but is not limited to, a mobile phone, a tablet computer, a notebook computer, an electric toy, an electric tool, an electric bicycle, an electric motorcycle, an electric vehicle, a ship,

or a spacecraft. The electric toy may include a fixed or movable electric toy, for example, a game console, an electric vehicle toy, an electric ship toy, or an electric aircraft toy, and the spacecraft may include an aircraft, a rocket, a space shuttle, or a spaceship.

[0118] For ease of description, description is provided by using an example in which an embodiment of the present application provides a power consuming device and the power consuming device is a vehicle.

[0119] Referring to FIG. 1, FIG. 1 is a schematic structural diagram of a vehicle **1000** according to some embodiments of the present application. The vehicle **1000** may be a fuel powered vehicle, a gas powered vehicle, or a new energy vehicle. The new energy vehicle may be a pure electric vehicle, a hybrid electric vehicle, or an extended range vehicle. A battery **1001** is arranged inside the vehicle **1000**, and the battery **1001** may be arranged at a bottom, head, or tail of the vehicle **1000**. The battery **1001** may be configured to supply power to the vehicle **1000**. For example, the battery **1001** may be used as an operating power supply of the vehicle **1000**. The vehicle **1000** may further include a controller **1002** and a motor **1003**. The controller **1002** is configured to control the battery **1001** to supply power to the motor **1003**, for example, to meet operating electricity requirements during starting, navigation, and traveling of the vehicle **1000**.

[0120] In some embodiments of the present application, the battery **1001** can not only be used as the operating power supply of the vehicle **1000**, but can also be used as a driving power supply of the vehicle **1000**, to alternatively or partially replace fuel or natural gas to provide driving power for the vehicle **1000**.

[0121] Referring to FIG. 2, FIG. 2 is a schematic exploded structural view of a battery **1001** according to some embodiments of the present application. The battery **1001** includes a box **100** and a battery cell **200**, where the battery cell **200** is accommodated in the box **100**. The box **100** is configured to provide an accommodating space for the battery cell **200**, and the box **100** may use a plurality of structures. In some embodiments, the box **100** may include a first part **101** and a second part **102**, the first part **101** and the second part **102** cover each other, and the first part **101** and the second part **102** jointly define the accommodating space for accommodating the battery cell **200**. The second part **102** may be a hollow structure being open on one side, the first part **101** may be a plate-like structure, and the first part **101** covers an open side of the second part **102**, so that the first part **101** and the second part **102** jointly define the accommodating space. Alternatively, each of the first part **101** and the second part **102** may be a hollow structure being open on one side, and an open side of the first part **101** covers an open side of the second part **102**. Certainly, the box **100** formed by the first part **101** and the second part **102** may be in various shapes, such as a cylinder or a cuboid. A plurality of battery cells **200** are connected in series, in parallel, or in series-parallel, and are then placed in the box **100** formed after the first part **101** and the second part **102** are buckled.

[0122] Referring to FIG. 3, FIG. 3 is a schematic exploded structural view of a battery cell **200** according to some embodiments of the present application. Further referring to FIG. 4 to FIG. 7, schematic structural diagrams of a connection part between an electrode terminal **41** and a wall portion **301** in a battery cell **200** according to some embodiments of the present application are shown. FIG. 4 is a schematic top structural view of mounting a wall portion **301** of an electrode terminal **41** on a housing according to some embodiments of the present application. FIG. 5 is a schematic cross-sectional structural view along a line A-A in FIG. 4. FIG. 6 is an enlarged view of a part B in FIG. 5. FIG. 7 is an enlarged view of a part C in FIG. 6.

[0123] An embodiment of the present application provides a battery cell **200**, including: an electrode assembly **21**; an electrode terminal **41**, electrically connected to the electrode assembly **21**; a housing **30**, configured to accommodate the electrode assembly **21**, where the housing **30** includes a wall portion **301** and a fixing structure **50**, and the fixing structure **50** is fixedly arranged on the wall portion **301** and configured to fix the electrode terminal **41**; an insulating member **42**, arranged between the fixing structure **50** and the electrode terminal **41**; and an insulating protection

member **60**, at least partially covering an outer surface of the fixing structure **50**, where the insulating protection member **60** includes an insertion portion **62**, and at least a part of the insertion portion **62** is inserted between the fixing structure **50** and the insulating member **42**.

[0124] The electrode terminal **41** is a conductive member arranged on the wall portion **301**, and the electrode terminal **41** is connected to a tab **211** of the electrode assembly **21**, to output electrical energy of the battery cell **200** or charge the battery cell **200**. Generally, the battery cell **200** includes two electrode terminals **41**, and the two electrode terminals **41** are respectively connected to positive and negative electrode tabs **211** of the electrode assembly **21**. That is, the housing **30** generally includes two electrode terminals **41**, the two electrode terminals **41** are respectively connected to two groups of tabs **211**, and the electrode terminals **41** are mounted on the wall portion **301** to support the electrode terminals **41** through the wall portion **301**. Certainly, the battery cell **200** may further include other functional components.

[0125] The housing **30** is a shell structure having an accommodating space inside, and the electrode assembly **21** is at least partially placed in the housing **30**, to protect the electrode assembly **21** through the housing **30**.

[0126] The housing **30** includes the wall portion **301**. The electrode terminal **41** is mounted on the wall portion **301** to support the electrode terminal **41** through the wall portion **301**, and the electrode terminal **41** is electrically connected to the tab **211**. The wall portion **301** is a partial region on the housing **30**.

[0127] The fixing structure **50** is a structure configured to fix the electrode terminal **41**. That the fixing structure **50** is fixedly arranged on the wall portion **301** refers to that the fixing structure **50** is connected to the wall portion **301** to support the fixing structure **50** through the wall portion **301**. The fixing structure **50** is fixedly connected to the electrode terminal **41**, to support the electrode terminal **41** through the fixing structure **50**, so that the electrode terminal **41** is fixed to the wall portion **301**.

[0128] The insulating member **42** is a component having an insulating characteristic. That the insulating member **42** is arranged between the fixing structure **50** and the electrode terminal **41** refers to that the insulating member **42** is arranged between the fixing structure **50** and the electrode terminal **41**, to insulate and isolate the fixing structure **50** from the electrode terminal **41** through the insulating member **42**. The insulating member **42** may be made of a material such as plastic, rubber, silicone, or bakelite that has an insulating characteristic.

[0129] The insulating protection member **60** is a structural member having an insulating characteristic. The insulating protection member **60** may be made of a material such as plastic, bakelite, or ceramic that has an insulating characteristic.

[0130] The outer surface of the fixing structure **50** is a surface of the fixing structure **50** facing away from the inside of the housing **30**.

[0131] During use, the outer surface of the fixing structure **50** is located on an outer side of the battery cell **200**, so that at least a part of the insulating protection member **60** covers the outer surface of the fixing structure **50**. Therefore, the outer surface of the fixing structure **50** can be well insulated and protected, thereby improving the insulating protection performance.

[0132] The insertion portion **62** is a structural part protruding from the insulating protection member **60** toward the inside of the housing **30**. Since the insulating member **42** is arranged between the fixing structure **50** and the electrode terminal **41**, by inserting at least a part of the insertion portion **62** between the fixing structure **50** and the insulating member **42**, a distance between the fixing structure **50** and the electrode terminal **41** can be increased, that is, a creepage distance between the electrode terminal **41** and the fixing structure **50** is increased, thereby improving the insulating safety. In addition, since at least a part of the insulating protection member **60** covers the outer surface of the fixing structure **50**, a side of the fixing structure **50** close to the electrode terminal **41** can be insulated and wrapped, to further improve an insulating protection effect, so that the fixing structure **50** is less likely to be in short circuit contact with the

electrode terminal **41**, to reduce a short circuit risk.

[0133] In the technical solution of the embodiments of the present application, the insulating protection member **60** is arranged to cover the outer surface of the fixing structure **50**, so that the outer surface of the fixing structure **50** is insulated and protected, and the insulation performance of the outer surface of the fixing structure **50** is improved, thereby reducing the influence of an external conductive member; and the insertion portion **62** is inserted between the fixing structure **50** and the insulating member **42**, so that the creepage distance between the electrode terminal **41** and the fixing structure **50** is increased, and the external conductive member can be better prevented from being inserted between the fixing structure **50** and the electrode terminal **41**, thereby improving the insulating safety and reducing a short circuit risk.

[0134] An inner surface of the wall portion **301** is a surface located inside the housing **30** on the wall portion **301**, and an outer surface of the wall portion **301** is a surface opposite to the inner surface on the wall portion **301**. The outer surface of the wall portion **301** and the inner surface of the wall portion **301** are two opposite surfaces of the wall portion **301**.

[0135] As shown in FIG. 3, the battery cell **200** has a height direction, a width direction, and a thickness direction. In the figure, a direction Z is the height direction of the battery cell **200**, a direction X is the width direction of the battery cell **200**, and a direction Y is the thickness direction of the battery cell **200**.

[0136] In some embodiments, the battery cell **200** may include one or more electrode assemblies **21**. When one electrode assembly **21** is used, assembly is facilitated. When a plurality of electrode assemblies **21** are used, a battery level capacity of the battery cell **200** can be improved.

[0137] In some embodiments, the electrode assembly **21** may be totally placed in the housing **30**, to well protect the electrode assembly **21** through the housing **30**.

[0138] In some embodiments, the electrode assembly **21** may alternatively be partially placed in the housing **30**. For example, when the electrode assembly **21** includes the tab **211**, the tab **211** may protrude from the housing **30**.

[0139] In some embodiments, the housing **30** may also be provided with a pressure relief mechanism configured to relieve internal pressure when the internal pressure or temperature in the battery cell **200** reaches a threshold.

[0140] In some embodiments, referring to FIG. 3, the battery cell **200** further includes two adapting pieces **22**, the two adapting pieces **22** respectively correspond to the two electrode terminals **41**, the adapting pieces **22** are connected to the corresponding electrode terminals **41**, and the adapting pieces **22** are connected to the corresponding tabs **211**. That is, the tabs **211** are connected to the corresponding electrode terminals **41** through the adapting pieces **22**, so that connection between the tabs **211** and the electrode terminals **41** is facilitated, and the connection is more stable.

[0141] In some embodiments, the battery cell **200** may include one or more electrode assemblies **21**. When one electrode assembly **21** is used, assembly is facilitated. When a plurality of electrode assemblies **21** are used, a battery level capacity of the battery cell **200** can be improved.

[0142] In some embodiments, the housing **30** includes a shell **31** and an end cap **32**, one end of the shell **31** has an opening **310**, the end cap **32** covers the opening **310**, the shell **31** includes a side wall **311** and a bottom wall **312**, the side wall **311** is annularly disposed on an outer side of the electrode assembly **21**, and the bottom wall **312** and the opening **310** are arranged opposite to each other; and the wall portion **301** is at least a partial region of the end cap **32**, or the wall portion **301** is at least a partial region of the bottom wall **312**, or the wall portion **301** is at least a partial region of the side wall **311**.

[0143] The end cap **32** is a component that covers the opening **310** of the shell **31** to isolate an internal environment of the battery cell **200** from an external environment. A shape of the end cap **32** may be adapted to a shape of the shell **31** to fit and cover the shell **31**. Optionally, the end cap **32** may be made of a material with specified hardness and strength (for example, aluminum alloy), so that the end cap **32** is less likely to deform under extrusion and collision, enabling the battery cell

200 to have a higher structural strength and improved safety performance.

[0144] The shell **31** is a component configured to form the internal environment of the battery cell **200** together with the end cap **32**, where the formed internal environment may be used for accommodating the electrode assembly **21**, an electrolyte, and other components. The shell **31** and the end cap **32** may be independent components, the opening **310** may be provided on the shell **31**, and the end cap **32** covers the opening **310** at the opening **310** to form the internal environment of the battery cell **200**. The shell **31** may be of various shapes and sizes, such as a rectangular shape, a cylindrical shape, or a hexagonal prism shape. Specifically, the shape of the shell **31** may be determined according to a specific shape and size of the battery cell **200**. The shell **31** may be made of various materials, and the materials include but are not limited to copper, iron, aluminum, stainless steel, aluminum alloy, or plastic.

[0145] The shell **31** includes the side wall **311** and the bottom wall **312**, the side wall **311** and the bottom wall **312** encircle to form an accommodating space to accommodate the electrode assembly **21**, and a side on the side wall **311** away from the bottom wall **312** forms the opening **310** of the shell **31** to place the electrode assembly **21**. After the electrode assembly **21** is mounted, the side wall **311** surrounds the electrode assembly **21**, the bottom wall **312** supports the electrode assembly **21**, and the end cap **32** covers the opening **310**.

[0146] The electrode terminal **41** may be mounted on the end cap **32**, and a partial region on the end cap **32** configured to mount the electrode terminal **41** is the wall portion **301** supporting the electrode terminal **41**. Certainly, the electrode terminal **41** may be mounted on the side wall **311**, and a partial region on the side wall **311** configured to mount the electrode terminal **41** is the wall portion **301** supporting the electrode terminal **41**. The electrode terminal **41** may alternatively be mounted on the bottom wall **312**, and a partial region on the bottom wall **312** configured to mount the electrode terminal **41** is the wall portion **301** supporting the electrode terminal **41**.

[0147] At least a partial region of the end cap **32** is used as the wall portion **301**, or at least a partial region of the bottom wall **312** is used as the wall portion **301**, or at least a partial region of the side wall **311** is used as the wall portion **301**, so that a layout design of the electrode terminal **41** is facilitated.

[0148] Since the electrode terminal **41** is mounted on the end cap **32**, manufacturing and assembly are relatively simple, mounting of the electrode terminal **41** is also facilitated, and connection between the electrode assembly **21** and the electrode terminal **41** is also facilitated. Detailed description is provided in the following embodiments by using an example in which a partial region of the end cap **32** is used as the wall portion **301** of the electrode terminal **41**.

[0149] When the insulating protection member **60** is mounted on the fixing structure **50**, a surface of the insulating protection member **60** close to the fixing structure **50** is generally viscous. For example, glue is coated on or an adhesive layer is arranged on the surface of the insulating protection member **60** close to the fixing structure **50**, so that the insulating protection member **60** is bonded to the fixing structure **50**, thereby preventing the insulating protection member **60** from falling off. However, since the insertion portion **62** is inserted between the fixing structure **50** and the insulating member **42**, when the insertion portion **62** is inserted between the fixing structure **50** and the insulating member **42**, the insertion portion encounters relatively large resistance, which is inconvenient for insertion, and a position of the insulating protection member **60** close to the insertion portion **62** may even tilt up. As a result, bonding between the insulating protection member **60** and the fixing structure **50** is not firm, and the quality of the battery cell **200** is affected.

[0150] In some embodiments, referring to FIG. 6 and FIG. 7, the insertion portion **62** has a non-sticking characteristic.

[0151] That the insertion portion **62** has a non-sticking characteristic refers to that the insertion portion **62** is not viscous, that is, no glue is coated on or no adhesive layer is arranged on the insertion portion **62**, and the insertion portion **62** is not bonded to the fixing structure **50** and the insulating member **42**. The insertion portion **62** has a non-sticking characteristic, so that the

insertion portion **62** can be inserted between the fixing structure **50** and the insulating member **42**, thereby facilitating assembly, and the insulating protection member **60** can better fit and cover the outer surface of the fixing structure **50**.

[0152] In some embodiments, referring to FIG. 6 and FIG. 7, the fixing structure **50** includes a first fixing structure **51** and a second fixing structure **52**, at least one of the first fixing structure **51** and the second fixing structure **52** is fixedly connected to the wall portion **301**, the first fixing structure **51** is located on an inner side of the second fixing structure **52** away from the housing **30** in a thickness direction of the wall portion **301**, the first fixing structure **51** cooperates with the second fixing structure **52** to clamp and fix the electrode terminal **41**, at least a part of the insulating protection member **60** covers an outer surface of the first fixing structure **51**, and at least a part of the insertion portion **62** is inserted between the first fixing structure **51** and the insulating member **42**.

[0153] The first fixing structure **51** is a structural component fixed and supported on the wall portion **301** and configured to fix and support the electrode terminal **41**.

[0154] The second fixing structure **52** is a structural component fixed and supported on the wall portion **301** and configured to cooperate with the first fixing structure **51** to fix and support the electrode terminal **41**.

[0155] That at least one of the first fixing structure **51** and the second fixing structure **52** is fixedly connected to the wall portion **301** refers to that the first fixing structure **51** is directly fixedly connected to the wall portion **301**, or the second fixing structure **52** is directly fixedly connected to the wall portion **301**, or both the first fixing structure **51** and the second fixing structure **52** are directly fixedly connected to the wall portion **301**, to support the first fixing structure **51** and the second fixing structure **52** through the wall portion **301**, so that the fixing structure **50** is fixed and supported.

[0156] The first fixing structure **51** is located on the inner side of the second fixing structure **52** away from the housing **30** in the thickness direction of the wall portion **301**, so that the outer surface of the first fixing structure **51** is the outer surface of the entire fixing structure **50**. That is, during use, the outer surface of the first fixing structure **51** is located on the outer side of the battery cell **200**, the outer surface of the first fixing structure **51** is prone to the influence of the external conductive member, leading to a risk that the first fixing structure is short circuited with the electrode terminal **41**.

[0157] The first fixing structure **51** cooperates with the second fixing structure **52** to clamp and fix the electrode terminal **41**. That is, the first fixing structure **51** and the second fixing structure **52** clamp a partial region of the electrode terminal **41** to fix the electrode terminal **41**, so as to fix the electrode terminal **41** on the wall portion **301**. The first fixing structure **51** and the second fixing structure **52** are arranged to clamp the electrode terminal **41** through cooperation, so that mounting and fixing of the electrode terminal **41** are facilitated, thereby facilitating assembly of the electrode terminal **41**.

[0158] At least a part of the insulating protection member **60** covers the outer surface of the first fixing structure **51**. Therefore, the outer surface of the first fixing structure **51** can be well insulated and protected, thereby improving the insulating protection performance. By inserting at least a part of the insertion portion **62** between the first fixing structure **51** and the insulating member **42**, a distance between the first fixing structure **51** and the electrode terminal **41** can be increased, that is, a creepage distance between the electrode terminal **41** and the first fixing structure **51** is increased, thereby improving the insulating safety. In addition, since at least a part of the insulating protection member **60** covers the outer surface of the first fixing structure **51**, a side of the first fixing structure **51** close to the electrode terminal **41** can be insulated and wrapped, to further improve the insulating protection performance, so that the first fixing structure **51** is less likely to be in short circuit contact with the electrode terminal **41**, to reduce a short circuit risk, thereby further improving the insulating protection performance of the entire battery cell **200**.

[0159] In some embodiments, for ease of mounting and fixing of the electrode terminal **41**, a flange portion **411** is arranged on a peripheral side of the electrode terminal **41**, and the flange portion **411** is a boss structure arranged on the peripheral side of the electrode terminal **41**. The flange portion **411** is arranged for ease of fixing to the wall portion **301**. By arranging the first fixing structure **51** and the second fixing structure **52**, the flange portion **411** on the peripheral side of the electrode terminal **41** may be clamped by the first fixing structure **51** and the second fixing structure **52** through cooperation, so that the electrode terminal **41** is fixed.

[0160] In some embodiments, the first fixing structure **51** may be set to an annular shape, to cooperate with the second fixing structure **52** to clamp and fix the electrode terminal **41**.

[0161] In some embodiments, the second fixing structure **52** may be set to an annular shape, to cooperate with the first fixing structure **51** to clamp and fix the electrode terminal **41**.

[0162] In some embodiments, the insulating member **42** may be arranged in cooperation with the peripheral side surrounding the electrode terminal **41**, to better insulate and isolate the electrode terminal **41** from the fixing structure **50** and the wall portion **301**, thereby improving the insulating protection performance. It may be understood that the insulating member **42** may alternatively be arranged only between the fixing structure **50** and the electrode terminal **41**.

[0163] In some embodiments, referring to FIG. 6 and FIG. 7, the first fixing structure **51** includes a connection portion **512** and a clamping portion **511**; and in the thickness direction of the wall portion **301**, the clamping portion **511** and the second fixing structure **52** are arranged opposite to each other, and the connection portion **512** is located between the clamping portion **511** and the second fixing structure **52**; the insulating member **42** includes a first insulating portion **421** and a second insulating portion **422**, at least a part of the first insulating portion **421** is located between the clamping portion **511** and the electrode terminal **41** in the thickness direction of the wall portion **301**, at least a part of the second insulating portion **422** is located between the clamping portion **511** and the electrode terminal **41** in a first direction, and the first direction is a direction pointing from the clamping portion **511** to the electrode terminal **41** and perpendicular to the thickness direction of the wall portion **301**; and at least a part of the insertion portion **62** is located between the first insulating portion **421** and the clamping portion **511**.

[0164] The clamping portion **511** is a partial structure that is on the first fixing structure **51** and cooperates with the second fixing structure **52** to clamp the electrode terminal **41**. The clamping portion **511** and the second fixing structure **52** are arranged opposite to each other, to clamp and fix the electrode terminal **41**. The connection portion **512** is a partial structure of the first fixing structure **51** for connecting the clamping portion **511** and the wall portion **301**, the clamping portion **511** is supported on the wall portion **301** through the connection portion **512**, and the connection portion **512** is located between the clamping portion **511** and the second fixing structure **52**, to fixedly connect the clamping portion **511** to the second fixing structure **52**.

[0165] The connection portion **512** is arranged in a manner of extending from the wall portion **301** toward a direction away from the inner surface of the wall portion **301**, the clamping portion **511** is arranged in a manner of extending from an end of the connection portion **512** away from the wall portion **301** toward a direction of the electrode terminal **41**, so that the connection portion **512** supports the clamping portion **511**, and the clamping portion **511** cooperates with the second fixing structure **52** to clamp the electrode terminal **41**.

[0166] The insulating member **42** includes the first insulating portion **421** and the second insulating portion **422**, and the first insulating portion **421** is a partial structure on the insulating member **42**. The second insulating portion **422** is a partial structure on the insulating member **42**.

[0167] That at least a part of the first insulating portion **421** is located between the clamping portion **511** and the electrode terminal **41** in the thickness direction of the wall portion **301** refers to that: the first insulating portion **421** totally or partially extends in the thickness direction of the wall portion **301** and is located between the clamping portion **511** and the electrode terminal **41**.

[0168] That at least a part of the second insulating portion **422** is located between the clamping

portion **511** and the electrode terminal **41** in a first direction refers to that: the second insulating portion **422** totally or partially extends in the first direction and is located between the clamping portion **511** and the electrode terminal **41**.

[0169] That the first direction is a direction pointing from the clamping portion **511** to the electrode terminal **41** and perpendicular to the thickness direction of the wall portion **301** refers to that: the first direction is a direction from the clamping portion **511** to the electrode terminal **41** and a direction perpendicular to the thickness direction of the wall portion **301**.

[0170] By arranging the first insulating portion **421**, the clamping portion **511** can be insulated and isolated from the electrode terminal **41** along an end surface in the first direction, and the second insulating portion **422** insulates and isolates a surface of the clamping portion **511** close to the second fixing structure **52** from the electrode terminal **41**, so that the electrode terminal is well insulated and protected.

[0171] That at least a part of the insertion portion **62** is located between the first insulating portion **421** and the clamping portion **511** refers to that: the insertion portion **62** totally or partially extends into a space between the first insulating portion **421** and the clamping portion **511**, to improve the insulating protection performance of the clamping portion **511**.

[0172] In some embodiments, referring to FIG. 6 and FIG. 7, the insertion portion **62** is provided with a first engagement portion **70**, a second engagement portion **80** is arranged between the first fixing structure **51** and the insulating member **42**, and the first engagement portion **70** is in engagement fit with the second engagement portion **80**.

[0173] The first engagement portion **70** is a structure portion arranged on the insertion portion **62** and configured for engagement.

[0174] The second engagement portion **80** is a structure portion arranged between the first fixing structure **51** and the insulating member **42** and configured for engagement with the first engagement portion **70**.

[0175] The first engagement portion **70** is arranged on the insertion portion **62**, the second engagement portion **80** is arranged between the first fixing structure **51** and the insulating member **42**, and the insertion portion **62** is fixed through engagement fit between the first engagement portion **70** and the second engagement portion **80**, so that the insertion portion **62** can be better prevented from falling off, and the insulating protection member **60** and the first fixing structure **51** can be better fitted and fixed.

[0176] In some embodiments, referring to FIG. 6 and FIG. 7, the first engagement portion **70** includes a limiting protrusion **71**, the second engagement portion **80** includes an accommodating gap **81** formed between the first fixing structure **51** and the second insulating portion **422**, and at least a part of the limiting protrusion **71** is accommodated in the accommodating gap **81**.

[0177] The limiting protrusion **71** is a protruding structure arranged on the insertion portion **62**, and as shown in FIG. 7, the limiting protrusion **71** may be a bump structure **711** protruding from the insertion portion **62**.

[0178] The accommodating gap **81** is a gap space formed between the first fixing structure **51** and the second insulating portion **422**.

[0179] That at least a part of the limiting protrusion **71** is accommodated in the accommodating gap **81** refers to that: the limiting protrusion **71** totally or partially extends into the accommodating gap **81**, to position the limiting protrusion **71**.

[0180] The accommodating gap **81** is provided between the first fixing structure **51** and the second insulating portion **422**, the limiting protrusion **71** is arranged on the insertion portion **62**, and the limiting protrusion **71** is accommodated through the accommodating gap **81**, so that the insertion portion **62** is clamped and positioned, and a structure is simple and easy to process and manufacture.

[0181] In some embodiments, referring to FIG. 6 and FIG. 7, the clamping portion **511** has a first end surface **5111** facing the first insulating portion **421**, the clamping portion **511** has a second end

surface **5112** facing the second insulating portion **422**, the first end surface **5111** and the second end surface **5112** are connected through a first transition surface **5113**, the first transition surface **5113** is a rounded surface **51131**, and the first transition surface **5113** and the second insulating portion **422** are spaced apart to form the accommodating gap **81**.

[0182] The first end surface **5111** is an end surface of the clamping portion **511** facing the first insulating portion **421**. The second end surface **5112** is an end surface of the clamping portion **511** facing the second insulating portion **422**.

[0183] The first transition surface **5113** is a surface of the clamping portion **511** connecting the first end surface **5111** and the second end surface **5112** in a transition manner.

[0184] That the first transition surface **5113** is a rounded surface **51131** refers to that the rounded surface **51131** is arranged between the first end surface **5111** and the second end surface **5112** on the clamping portion **511**, to form the first transition surface **5113**. Due to arrangement of the rounded surface **51131**, the rounded surface **51131** is isolated from the second insulating portion **422**, so that the rounded surface **51131** and the second insulating portion **422** are spaced apart to form the accommodating gap **81**. In this way, the limiting protrusion **71** may partially or totally extend into the accommodating gap **81**, to stop and position the limiting protrusion **71** through the rounded surface **51131**.

[0185] The rounded surface **51131** is arranged on the clamping portion **511** to form the first transition surface **5113**, so that a structure is simple and easy to manufacture.

[0186] In some embodiments, referring to FIG. **8**, the clamping portion **511** has a first end surface **5111** facing the first insulating portion **421**, the clamping portion **511** has a second end surface **5112** facing the second insulating portion **422**, the first end surface **5111** and the second end surface **5112** are connected through a first transition surface **5113**, the first transition surface **5113** is a chamfered surface **51132**, and the first transition surface **5113** and the second insulating portion **422** are spaced apart to form the accommodating gap **81**.

[0187] That the first transition surface **5113** is a chamfered surface **51132** refers to that the chamfered surface **51132** is arranged between the first end surface **5111** and the second end surface **5112** on the clamping portion **511**, to form the first transition surface **5113**. Due to arrangement of the chamfered surface **51132**, the chamfered surface **51132** is isolated from the second insulating portion **422**, so that the chamfered surface **51132** and the second insulating portion **422** are spaced apart to form the accommodating gap **81**. In this way, the limiting protrusion **71** may partially or totally extend into the accommodating gap **81**, to stop and position the limiting protrusion **71** through the chamfered surface **51132**.

[0188] The chamfered surface **51132** is arranged on the clamping portion **511** to form the first transition surface **5113**, so that a structure is simple and easy to manufacture.

[0189] In some embodiments, the clamping portion **511** and the second insulating portion **422** may alternatively be spaced apart to form the accommodating gap **81**, so as to accommodate the limiting protrusion **71**.

[0190] In some embodiments, referring to FIG. **9** and FIG. **10**, the first engagement portion **70** includes a limiting protrusion **71**, the second engagement portion **80** is arranged on at least one of the first fixing structure **51** and the insulating member **42**, the second engagement portion **80** includes a limiting groove **82**, and at least a part of the limiting protrusion **71** is accommodated in the limiting groove **82**.

[0191] The limiting protrusion **71** is a protruding structure arranged on the insertion portion **62**, and as shown in FIG. **9**, the limiting protrusion **71** may be a hook **712** arranged on the insertion portion **62** in a protruding manner.

[0192] The limiting groove **82** is a groove body structure arranged on at least one of the first fixing structure **51** and the insulating member **42**. Referring to FIG. **9**, the limiting groove **82** may be a recessed groove **821** provided on the first fixing structure **51**. Referring to FIG. **10**, the limiting groove **82** may alternatively be a clamping slot **822** provided on the first fixing structure **51**.

Certainly, the limiting groove **82** may alternatively be provided on the insulating member **42**, or the limiting groove **82** is provided on both the insulating member **42** and the first fixing structure **51**. [0193] A position of the limiting protrusion **71** corresponds to the limiting groove **82**. For example, the limiting groove **82** is provided on the first fixing structure **51**, and the limiting protrusion **71** on the insertion portion **62** is located on a side of the insertion portion **62** close to the first fixing structure **51**, to extend into the limiting groove **82** on the first fixing structure **51**. For another example, the limiting groove **82** is provided on the insulating member **42**, and the limiting protrusion **71** on the insertion portion **62** is located on a side of the insertion portion **62** close to the insulating member **42**, to extend into the limiting groove **82** on the first fixing structure **51**. For another example, when the limiting groove **82** is provided on both the insulating member **42** and the first fixing structure **51**, the limiting protrusion **71** may be arranged on the side of the insertion portion **62** close to the first fixing structure **51** and the side of the insertion portion close to the insulating member **42**.

[0194] The limiting groove **82** is provided on at least one of the first fixing structure **51** and the insulating member **42**, the limiting protrusion **71** is arranged on the insertion portion **62**, and the limiting protrusion **71** is accommodated through the limiting groove **82**, so that the insertion portion **62** is clamped and fixed, and a structure is simple and easy to process and manufacture.

[0195] In some embodiments, referring to FIG. **6**, the insulating protection member **60** includes a first coverage portion **61** covering an outer surface of the clamping portion **511**, and the insertion portion **62** is connected to the first coverage portion **61**.

[0196] The outer surface of the clamping portion **511** is a surface of a side of the clamping portion **511** away from the second fixing structure **52**.

[0197] The first coverage portion **61** is a partial structure covering the outer surface of the clamping portion **511**. The insertion portion **62** is connected to the first coverage portion **61**, to support the insertion portion **62** through the first coverage portion **61**.

[0198] The first coverage portion **61** is arranged to cover the outer surface of the clamping portion **511**, and the insertion portion **62** is connected to the first coverage portion **61**, so that the outer surface of the clamping portion **511** and a surface close to the electrode terminal **41** are well insulated and protected, thereby improving the insulation performance.

[0199] Referring to FIG. **11** and FIG. **12**, FIG. **11** is a schematic cross-sectional structural view of a connection part between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application. FIG. **12** is an enlarged view of a part D in FIG. **11**.

[0200] In some embodiments, the first coverage portion **61** is bonded to the clamping portion **511**.

[0201] That the first coverage portion **61** is bonded to the clamping portion **511** refers to that the first coverage portion **61** is fixedly connected to the clamping portion **511** in a bonding manner.

[0202] The first coverage portion **61** is bonded to the clamping portion **511**, so that the clamping portion **511** is insulated and protected, and the connection and fixing between the first coverage portion **61** and the clamping portion **511** are facilitated, thereby facilitating connection and fixing between the insulating protection member **60** and the first fixing structure **51**.

[0203] In some embodiments, a first adhesive layer **91** may be arranged on a surface of the first coverage portion **61** close to the clamping portion **511**, so that the first coverage portion **61** is bonded to the outer surface of the clamping portion **511** through the first adhesive layer **91**, thereby facilitating assembly.

[0204] In some embodiments, an adhesive layer may alternatively be arranged and coated on a side of the first coverage portion **61** close to the outer surface of the clamping portion **511**, so that the first coverage portion is bonded to the outer surface of the clamping portion **511**, thereby facilitating assembly.

[0205] In some embodiments, the first coverage portion **61** may alternatively be welded to the outer surface of the clamping portion **511**. For example, the first coverage portion **61** is welded to the outer surface of the clamping portion **511** in an ultrasonic welding manner, to ensure that the first

coverage portion **61** is firmly connected to the clamping portion **511**.

[0206] In some embodiments, an engagement structure may be further arranged on the first coverage portion **61**, and a matching buckling structure is arranged on the clamping portion **511**, so that the first coverage portion **61** is connected to the clamping portion **511** in an engagement manner. For example, a bulge may be arranged on the first coverage portion **61**, and a latching groove is provided on the clamping portion **511** for clamping and connection.

[0207] In some embodiments, referring to FIG. **11**, the insulating protection member **60** further includes a second coverage portion **63** attached against an outer surface of the connection portion **512**, and the second coverage portion **63** is connected to the first coverage portion **61**.

[0208] The outer surface of the connection portion **512** is a surface of a side of the connection portion **512** away from the electrode terminal **41**.

[0209] The second coverage portion **63** is a partial region corresponding to the connection portion **512** on the insulating protection member **60**. The second coverage portion **63** is arranged and attached against the outer surface of the connection portion **512**, so that the connection portion **512** can be insulated and protected, thereby better improving the insulating protection performance. In addition, the arrangement of the second coverage portion **63** can cooperate with the insertion portion **62**, so that the second coverage portion is better connected and fixed to the first fixing structure **51**.

[0210] In some embodiments, a second transition surface **513** is arranged between the outer surface of the clamping portion **511** and the outer surface of the connection portion **512**, and the second transition surface **513** may be an inclined surface or may be an arc surface, thereby facilitating processing and manufacturing of the first fixing structure **51**.

[0211] In some embodiments, when the first fixing structure **51** includes the second transition surface **513**, the insulating protection member **60** further includes a first transition portion **64**, the first transition portion **64** connects the first coverage portion **61** to the second coverage portion **63**, and a shape of the first transition portion **64** matches a shape of the second transition surface **513**, so that the first transition portion **64** is attached against the second transition surface **513**, thereby insulating and protecting the second transition surface **513**.

[0212] In some embodiments, a surface of the second coverage portion **63** close to the connection portion **512** has a non-sticking characteristic.

[0213] That a surface of the second coverage portion **63** close to the connection portion **512** has a non-sticking characteristic refers to that the surface of the second coverage portion **63** close to the connection portion **512** is not viscous. That is, no glue is coated on or no adhesive layer is arranged on the surface of the second coverage portion **63** close to the connection portion **512**, and the surface of the second coverage portion **63** close to the connection portion **512** is not bonded to the connection portion **512**. The surface of the second coverage portion **63** close to the connection portion **512** has the non-sticking characteristic, so that matching and connection between the second coverage portion **63** and the connection portion **512** are facilitated for ease of assembly, and the insulating protection member **60** can better fit and cover the outer surface of the first fixing structure **51**.

[0214] Referring to FIG. **13**, FIG. **13** is a schematic cross-sectional structural view of a connection portion **512** between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application.

[0215] In some embodiments, a first clamping protrusion **631** is arranged on the second coverage portion **63**, and a first clamping groove **5121** that is engaged with the first clamping protrusion **631** is provided on the connection portion **512**.

[0216] The first clamping protrusion **631** is a protruding structure arranged on the second coverage portion **63**. The first clamping protrusion **631** is arranged in a manner of protruding from the second coverage portion **63** toward the connection portion **512**. The first clamping groove **5121** is a groove structure arranged on the connection portion **512**. The first clamping groove **5121** is formed

by a surface of the connection portion **512** away from the electrode terminal **41** recessing inward. The first clamping groove **5121** matches the first clamping protrusion **631**.

[0217] The first clamping protrusion **631** is arranged on the second coverage portion **63**, and the first clamping groove **5121** is provided on the connection portion **512**, so that the second coverage portion **63** can be fixedly connected to the connection portion **512** through engagement between the first clamping protrusion **631** and the first clamping groove **5121**, and the second coverage portion is further engaged with the insertion portion **62**, to fix the insulating protection member **60** on the first fixing structure **51**, thereby facilitating assembly and fixing.

[0218] In some embodiments, a partial region on the second coverage portion **63** can protrude or bend toward the connection portion **512** to form the first clamping protrusion **631**. Certainly, the first clamping protrusion **631** may alternatively be arranged on the second coverage portion **63**.

[0219] Referring to FIG. **14** and FIG. **15**, FIG. **14** is a schematic cross-sectional structural view of a connection part between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application. FIG. **15** is an enlarged view of a part E in FIG. **14**.

[0220] In some embodiments, a first engagement groove **632** may be provided on the second coverage portion **63**, and a first engagement protrusion **5122** that is engaged with the first engagement groove **632** is arranged on the connection portion **512**.

[0221] The first engagement groove **632** is a groove structure arranged on the second coverage portion **63**. The first engagement protrusion **5122** is a protruding structure arranged on the connection portion **512**. The first engagement groove **632** matches the first engagement protrusion **5122**.

[0222] The first engagement groove **632** is provided on the second coverage portion **63**, and the first engagement protrusion **5122** is arranged on the connection portion **512**, so that the second coverage portion **63** can be fixedly connected to the connection portion **512** through engagement between the first engagement protrusion **5122** and the first engagement groove **632**, and the second coverage portion is further engaged with the insertion portion **62**, to fix the insulating protection member **60** on the first fixing structure **51**, thereby facilitating assembly and fixing.

[0223] In some embodiments, both the first clamping protrusion **631** and the first engagement groove **632** may be provided on the second coverage portion **63**, the first clamping groove **5121** that is engaged with the first clamping protrusion **631** is provided on the connection portion **512**, and the first engagement protrusion **5122** that is engaged with the first engagement groove **632** is arranged on the connection portion **512**.

[0224] Referring to FIG. **16**, FIG. **16** is a schematic cross-sectional structural view of a connection portion between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application.

[0225] In some embodiments, an angle α between the second coverage portion **63** and the first coverage portion **61** is an acute angle.

[0226] The angle α between the second coverage portion **63** and the first coverage portion **61** is an angle between a surface on which the second coverage portion **63** is located and a surface on which the first coverage portion **61** is located. The angle α between the second coverage portion **63** and the first coverage portion **61** is an acute angle, so that the second coverage portion **63** can cooperate with the insertion portion **62** to clamp the first fixing structure **51**, and the insulating protection member **60** is further connected to the first fixing structure **51**, thereby facilitating assembly.

[0227] In some embodiments, the second coverage portion **63** may be arranged in a manner of extending toward the electrode terminal **41** in a direction from the clamping portion **511** to the second fixing structure **52**, so that the angle α between the second coverage portion **63** and the first coverage portion **61** forms an acute angle.

[0228] In some embodiments, the surface of the connection portion **512** away from the electrode terminal **41** is recessed inward to match the second coverage portion **63**, so that the second coverage portion **63** is in engagement fit with the connection portion **512**, and the insulating

protection member **60** is better fixed onto the first fixing structure **51**.

[0229] Referring to FIG. **17**, FIG. **17** is a schematic cross-sectional structural view of a connection portion between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application.

[0230] In some embodiments, the insulating protection member **60** further includes an extension portion **66** covering an outer surface of the wall portion **301**, and the extension portion **66** is connected to the second coverage portion **63**.

[0231] The extension portion **66** is a partial structure on the insulating protection member **60** extending from the second coverage portion **63** to the outer surface of the wall portion **301**.

[0232] The extension portion **66** is arranged to extend from the second coverage portion **63** to the outer surface of the wall portion **301**, so that an insulating protection effect can be better achieved.

[0233] In some embodiments, a third transition surface **514** is arranged between the outer surface of the connection portion **512** and the outer surface of the wall portion **301**, and the third transition surface **514** may be an inclined surface or may be an arc surface, thereby facilitating processing and manufacturing of the first fixing structure **51** and connection and fixing between the first fixing structure **51** and the wall portion **301**.

[0234] In some embodiments, when the first fixing structure **51** includes the third transition surface **514**, the insulating protection member **60** further includes a second transition portion **65**, the second transition portion **65** is connected to the second coverage portion **63**, and a shape of the second transition portion **65** matches a shape of the third transition surface **514**, so that the second transition portion **65** is attached against the third transition surface **514**, thereby insulating and protecting the third transition surface **514**.

[0235] In some embodiments, when the insulating protection member **60** includes the extension portion **66** and the second transition portion **65**, the extension portion **66** is connected to the second transition portion **65**. That is, the second transition portion **65** is located between the extension portion **66** and the second coverage portion **63**, and the second transition portion **65** connects the extension portion **66** to the second coverage portion **63**, so that the insulating protection member **60** can well cooperate with the first fixing structure **51**, and the extension portion **66** is transitioned and attached to the wall portion **301**.

[0236] Referring to FIG. **18** and FIG. **19**, FIG. **18** is a schematic cross-sectional structural view of a connection part between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application. FIG. **19** is an enlarged view of a part F in FIG. **18**.

[0237] In some embodiments, a second clamping protrusion **661** is arranged on the extension portion **66**, and a second clamping groove **3011** that is engaged with the second clamping protrusion **661** is provided on the wall portion **301**.

[0238] The second clamping protrusion **661** is a protruding structure arranged on the extension portion **66**. The second clamping protrusion **661** is arranged in a manner of protruding from the extension portion **66** toward the wall portion **301**. The second clamping groove **3011** is a groove structure arranged on the wall portion **301**. The second clamping groove **3011** is formed by the outer surface of the wall portion **301** recessing inward. The second clamping groove **3011** matches the second clamping protrusion **661**.

[0239] The second clamping protrusion **661** is arranged on the extension portion **66**, and the second clamping groove **3011** is provided on the wall portion **301**, so that the extension portion **66** is fixedly connected to the wall portion **301** through engagement between the second clamping protrusion **661** and the second clamping groove **3011**, and the extension portion is further engaged with the insertion portion **62**, to fix the insulating protection member **60**, thereby facilitating assembly and fixing.

[0240] In some embodiments, a partial region on the extension portion **66** can protrude or bend toward the wall portion **301** to form the second clamping protrusion **661**. Certainly, the second clamping protrusion **661** may alternatively be arranged on the extension portion **66**.

[0241] Referring to FIG. 20, FIG. 20 is a schematic structural diagram of connection between an extension portion **66** and a wall portion **301** on a housing **30** according to some embodiments of the present application.

[0242] In some embodiments, a second engagement groove **662** is provided on the extension portion **66**, and a second engagement protrusion **3012** that is engaged with the second engagement groove **662** is arranged on the wall portion **301**.

[0243] The second engagement groove **662** is a groove structure arranged on the extension portion **66**. The second engagement protrusion **3012** is a protruding structure arranged on the wall portion **301**. The second engagement groove **662** matches the second engagement protrusion **3012**.

[0244] The second engagement groove **662** is provided on the extension portion **66**, and the second engagement protrusion **3012** is arranged on the wall portion **301**, so that the extension portion **66** is fixedly connected to the wall portion **301** through engagement between the second engagement protrusion **3012** and the second engagement groove **662**, and the extension portion is further engaged with the insertion portion **62**, to fix the insulating protection member **60**, thereby facilitating assembly and fixing.

[0245] In some embodiments, both the second clamping protrusion **661** and the second engagement groove **662** are provided on the extension portion **66**, the second clamping groove **3011** that is engaged with the second clamping protrusion **661** is provided on the wall portion **301**, and the second engagement protrusion **3012** that is engaged with the second engagement groove **662** is arranged on the wall portion **301**.

[0246] In some embodiments, the extension portion **66** covers the outer surface of the wall portion **301**. The extension portion **66** covers the outer surface of the wall portion **301**, to well protect the wall portion **301**, thereby improving the insulating protection performance of the wall portion **301**. It may be understood that, in some embodiments, the extension portion **66** may alternatively cover a partial region of the outer surface of the wall portion **301**.

[0247] Referring to FIG. 21 and FIG. 22, FIG. 21 is a schematic cross-sectional structural view of a connection part between an electrode terminal **41** and a wall portion **301** on a housing **30** according to some embodiments of the present application. FIG. 22 is an enlarged view of a part G in FIG. 21.

[0248] In some embodiments, the extension portion **66** is bonded to the wall portion **301**.

[0249] That the extension portion **66** is bonded to the wall portion **301** refers to that the extension portion **66** is fixedly connected to the wall portion **301** in a bonding manner.

[0250] The extension portion **66** is bonded to the wall portion **301**, so that the extension portion **66** is fixedly connected to the wall portion **301**, and the extension portion is further engaged with the insertion portion **62**, to fix the insulating protection member **60**, thereby facilitating assembly and fixing.

[0251] In some embodiments, a second adhesive layer **92** may be arranged on an outer surface of the extension portion **66** close to the wall portion **301**, so that the extension portion **66** is bonded to the outer surface of the wall portion **301** through the second adhesive layer **92**, thereby facilitating assembly.

[0252] In some embodiments, an adhesive layer may alternatively be arranged and coated on a surface of the extension portion **66** close to the wall portion **301**, so that the extension portion is bonded to the outer surface of the wall portion **301**, thereby facilitating assembly.

[0253] In some embodiments, the extension portion **66** may alternatively be welded to the outer surface of the wall portion **301**. For example, the extension portion **66** is welded to the outer surface of the wall portion **301** in an ultrasonic welding manner, to ensure that the extension portion **66** is firmly connected to the wall portion **301**.

[0254] In some embodiments, the first fixing structure **51** is an integrally formed structure, thereby facilitating processing and manufacturing of the first fixing structure **51**.

[0255] In some embodiments, the first coverage portion **61**, the second coverage portion **63**, the

extension portion **66**, and the insertion portion **62** are integrally formed and arranged.

[0256] The first coverage portion **61**, the second coverage portion **63**, the extension portion **66**, and the insertion portion **62** are integrally formed, so that the insulating protection member **60** is integrally formed, to facilitate processing and manufacturing of the insulating protection member **60**.

[0257] In some embodiments, referring to FIG. **6** again, the first fixing structure **51** and the wall portion **301** are an integrally formed structure, that is, the first fixing structure **51** and the wall portion **301** are integrally manufactured. The first fixing structure **51** and the wall portion **301** are integrally formed, so that connection and fixing between the first fixing structure **51** and the wall portion **301** are facilitated, and processing and manufacturing are also facilitated.

[0258] In some embodiments, referring to FIG. **11** again, the first fixing structure **51** and the second fixing structure **52** are an integrally formed structure, that is, the first fixing structure **51** and the second fixing structure **52** are integrally manufactured. In other words, the fixing structure **50** is an integrally formed structure. The first fixing structure **51** and the second fixing structure **52** are integrally formed, so that connection and fixing between the first fixing structure **51** and the second fixing structure **52** are facilitated, and processing and manufacturing are also facilitated.

[0259] In some embodiments, referring to FIG. **13** again, the second fixing structure **52** and the wall portion **301** are an integrally formed structure, that is, the second fixing structure **52** and the wall portion **301** are integrally manufactured. The second fixing structure **52** and the wall portion **301** are integrally formed, so that connection and fixing between the second fixing structure **52** and the wall portion **301** are facilitated, and processing and manufacturing are also facilitated.

[0260] In some embodiments, referring to FIG. **6** again, the second fixing structure **52** may be independently manufactured and then fixedly connected to the wall portion **301**, so that the second fixing structure **52** cooperates with the first fixing structure **51** to clamp the electrode terminal **41**, thereby facilitating assembly.

[0261] In some embodiments, referring to FIG. **11** again, the fixing structure **50** is integrally formed, and the fixing structure **50** is independently manufactured and then fixedly connected to the wall portion **301**, thereby facilitating processing and manufacturing.

[0262] In some embodiments, referring to FIG. **13** again, the first fixing structure **51** may be independently manufactured and then fixedly connected to the wall portion **301**, so that the first fixing structure **51** cooperates with the second fixing structure **52** to clamp the electrode terminal **41**, thereby facilitating assembly.

[0263] In some embodiments, referring to FIG. **16**, the fixing structure **50** and the wall portion **301** are an integrally formed structure, that is, the fixing structure **50** and the wall portion **301** are integrally manufactured. In other words, the first fixing structure **51**, the second fixing structure **52**, and the wall portion **301** are integrally formed. The fixing structure **50** and the wall portion **301** are integrally formed, so that connection and fixing between the fixing structure **50** and the wall portion **301** are facilitated, and processing and manufacturing are also facilitated.

[0264] In some embodiments, an insulating sealing member **43** is further arranged between the fixing structure **50** and the electrode terminal **41**.

[0265] The insulating sealing member **43** is a sealing structural member having an insulating characteristic. The insulating sealing member **43** may be made of a material such as silicone or rubber that has an insulating characteristic.

[0266] The insulating sealing member **43** is arranged between the fixing structure **50** and the electrode terminal **41**, so that the sealing performance is improved, and the insulating protection performance between the fixing structure **50** and the electrode terminal **41** can be improved.

[0267] In some embodiments, the fixing structure **50** includes the first fixing structure **51** and the second fixing structure **52**, the insulating sealing member **43** is arranged between the second fixing structure **52** and the electrode terminal **41**, and the insulating member **42** is arranged between the first fixing structure **51** and the electrode terminal **41**, so that the insulating sealing member **43**

cooperates with the insulating member **42**, and the electrode terminal **41** is insulated and isolated from the fixing structure **50**.

[0268] A battery cell **200** is provided according to some embodiments of the present application, including a housing **30**, an electrode assembly **21**, an electrode terminal **41**, an insulating member **42**, and an insulating sealing member **43**. The housing **30** includes a wall portion **301** and a fixing structure **50**, the fixing structure **50** includes a first fixing structure **51** and a second fixing structure **52**, at least one of the first fixing structure **51** and the second fixing structure **52** is fixedly connected to the wall portion **301**, the first fixing structure **51** is located on an inner side of the second fixing structure **52** away from the housing **30**, and the first fixing structure **51** cooperates with the second fixing structure **52** to clamp and fix the electrode terminal **41**, to fix the electrode terminal **41** onto the wall portion **301**. The insulating sealing member **43** is arranged between the second fixing structure **52** and the electrode terminal **41**, and insulating member **42** is arranged between the first fixing structure **51** and the electrode terminal **41**, so that the insulating sealing member **43** cooperates with the insulating member **42**, to insulate and isolate the electrode terminal **41** from the fixing structure **50**. The insulating protection member **60** at least partially covers an outer surface of the first fixing structure **51**, the insulating protection member **60** includes an insertion portion **62**, and at least a part of the insertion portion **62** is inserted between the first fixing structure **51** and the insulating member **42**. The insulation performance of an outer surface of the fixing structure **50** is improved, thereby reducing the influence of an external conductive member and reducing a short circuit risk. The insertion portion **62** has a non-sticking characteristic, so that the insertion portion **62** can be inserted between the fixing structure **50** and the insulating member **42**, thereby facilitating assembly, and the insulating protection member **60** can better fit and cover the outer surface of the fixing structure **50**.

[0269] According to some embodiments of the present application, the present application further provides a battery, including the battery cell **200** according to any one of the foregoing solutions.

[0270] According to some embodiments of the present application, the present application further provides a power consuming device, including the battery according to any one of the foregoing solutions.

[0271] The power consuming device may be any device or system in which the foregoing battery is used.

[0272] Finally, it should be noted that the foregoing embodiments are only used for describing the technical solutions of the present application, but are not intended to limit the present application. Although the present application has been described in detail with reference to the foregoing embodiments, a person of ordinary skill in the art should appreciate that they can still make modifications to the technical solutions recorded in the embodiments or make equivalent replacements to some or all technical features of the present application. These modifications or replacements do not cause the essence of corresponding technical solutions to depart from the scope of the technical solutions in the embodiments of the present application, and shall all fall within the scope of the claims and specification of the present application. Especially, as long as there is no structural conflict, the various technical features mentioned in each embodiment may be combined in any way. The present application is not limited to the particular embodiments disclosed herein, but includes all technical solutions that fall within the scope of the claims.

Claims

1. A battery cell, comprising: an electrode assembly; an electrode terminal, electrically connected to the electrode assembly; a housing, configured to accommodate the electrode assembly, wherein the housing comprises a wall portion and a fixing structure, and the fixing structure is fixedly arranged on the wall portion and configured to fix the electrode terminal; an insulating member, arranged between the fixing structure and the electrode terminal; and an insulating protection member, at

least partially covering an outer surface of the fixing structure, wherein the insulating protection member comprises an insertion portion, and at least a part of the insertion portion is inserted between the fixing structure and the insulating member.

2. The battery cell according to claim 1, wherein the fixing structure comprises a first fixing structure and a second fixing structure, at least one of the first fixing structure and the second fixing structure is fixedly connected to the wall portion, the first fixing structure is located on an inner side of the second fixing structure away from the housing in a thickness direction of the wall portion, the first fixing structure cooperates with the second fixing structure to clamp and fix the electrode terminal, at least a part of the insulating protection member covers an outer surface of the first fixing structure, and at least a part of the insertion portion is inserted between the first fixing structure and the insulating member.

3. The battery cell according to claim 2, wherein the first fixing structure comprises a connection portion and a clamping portion; in the thickness direction of the wall portion, the clamping portion and the second fixing structure are arranged opposite to each other, and the connection portion is located between the clamping portion and the second fixing structure; the insulating member comprises a first insulating portion and a second insulating portion, at least a part of the first insulating portion is located between the clamping portion and the electrode terminal in the thickness direction of the wall portion, at least a part of the second insulating portion is located between the clamping portion and the electrode terminal in a first direction, and the first direction is a direction pointing from the clamping portion to the electrode terminal and perpendicular to the thickness direction of the wall portion; and at least a part of the insertion portion is located between the first insulating portion and the clamping portion.

4. The battery cell according to claim 3, wherein the insertion portion is provided with a first engagement portion, a second engagement portion is arranged between the first fixing structure and the insulating member, and the first engagement portion is in engagement fit with the second engagement portion.

5. The battery cell according to claim 4, wherein the first engagement portion comprises a limiting protrusion, the second engagement portion is arranged on at least one of the first fixing structure and the insulating member, the second engagement portion comprises a limiting groove, and at least a part of the limiting protrusion is accommodated in the limiting groove.

6. The battery cell according to claim 4, wherein the first engagement portion comprises a limiting protrusion, the second engagement portion comprises an accommodating gap formed between the first fixing structure and the second insulating portion, and at least a part of the limiting protrusion is accommodated in the accommodating gap.

7. The battery cell according to claim 6, wherein the clamping portion has a first end surface facing the first insulating portion, the clamping portion has a second end surface facing the second insulating portion, the first end surface and the second end surface are connected through a first transition surface, the first transition surface is a rounded surface or a chamfered surface, and the first transition surface and the second insulating portion are spaced apart to form the accommodating gap.

8. The battery cell according to claim 3, wherein the insulating protection member comprises a first coverage portion covering an outer surface of the clamping portion, and the insertion portion is connected to the first coverage portion.

9. The battery cell according to claim 8, wherein the first coverage portion is bonded to the clamping portion.

10. The battery cell according to claim 8, wherein the insulating protection member further comprises a second coverage portion attached against an outer surface of the connection portion, and the second coverage portion is connected to the first coverage portion.

11. The battery cell according to claim 10, wherein a first clamping protrusion is arranged on the second coverage portion, and a first clamping groove that is engaged with the first clamping

protrusion is provided on the connection portion; and/or a first engagement groove is provided on the second coverage portion, and a first engagement protrusion that is engaged with the first engagement groove is arranged on the connection portion.

12. The battery cell according to claim 10, wherein an angle between the second coverage portion and the first coverage portion is an acute angle.

13. The battery cell according to claim 10, wherein a surface of the second coverage portion close to the connection portion has a non-sticking characteristic.

14. The battery cell according to claim 10, wherein the insulating protection member further comprises an extension portion covering an outer surface of the wall portion, and the extension portion is connected to the second coverage portion.

15. The battery cell according to claim 14, wherein the first coverage portion, the second coverage portion, the extension portion, and the insertion portion are integrally formed and arranged.

16. The battery cell according to claim 2, wherein the first fixing structure and the wall portion are an integrally formed structure; and/or the second fixing structure and the wall portion are an integrally formed structure; and/or the first fixing structure and the second fixing structure are an integrally formed structure.

17. The battery cell according to claim 1, wherein an insulating sealing member is further arranged between the fixing structure and the electrode terminal.

18. The battery cell according to claim 1, wherein the insertion portion has a non-sticking characteristic.

19. The battery cell according to claim 1, wherein the housing comprises a shell and an end cap, one end of the shell has an opening, the end cap covers the opening, the shell comprises a side wall and a bottom wall, the side wall is annularly disposed on an outer side of the electrode assembly, and the bottom wall and the opening are arranged opposite to each other; and the wall portion is at least a partial region of the end cap, or the wall portion is at least a partial region of the bottom wall, or the wall portion is at least a partial region of the side wall.

20. A battery, comprising the battery cell according to claim 1.
